SPH4U Grade 12 Physics - Unit 1 Kinematics Practice Test (Version 1)

Name: _____ **Date:** _____ **Total:** 60 marks

This paper covers the Unit 1 syllabus and Nelson Physics 12 Chapter 1. Show complete work in Sections C and D. Use **g = 9.80 m/s² downward** and neglect air resistance where a projectile is involved. Give directions for vector answers. A calculator and graph paper may be used.

A. Multiple Choice (15 marks)

Choose the **one best answer**. Each question is worth 1 mark.

1. A student walks 30 m east and then 30 m west, returning to the start. The student's average velocity for the whole walk is:

A. 0 m/s B. 30 m/s east C. 60 m/s east D. impossible to determine because time is missing

2. The slope of a position-time graph at one instant gives the object's:

A. displacement B. instantaneous velocity C. acceleration D. distance travelled

3. The signed area between a velocity-time graph and the time axis gives:

A. acceleration B. average speed C. displacement D. final position in every case

4. A speed of 72 km/h is equal to:

A. 2.0 m/s B. 20 m/s C. 72 m/s D. 259 m/s

5. At the highest point of a projectile's flight, its acceleration is:

A. zero B. horizontal C. 9.80 m/s² downward D. 9.80 m/s² upward

6. An object's velocity changes uniformly from +4.0 m/s to -2.0 m/s in 3.0 s. Its displacement during the interval is:

A. -3.0 m B. 0 m C. +3.0 m D. +9.0 m

7. A displacement of 6.0 m east followed by 8.0 m north has a resultant of:

A. 10 m [53° N of E] B. 10 m [37° N of E] C. 14 m [53° N of E] D. 2.0 m [N]

8. A puck's velocity changes from 4.0 m/s east to 3.0 m/s north. Its change in velocity is:

A. 5.0 m/s [37° N of W] B. 5.0 m/s [37° N of E] C. 7.0 m/s [N] D. 1.0 m/s [W]

9. A ball is launched horizontally at 12 m/s and is airborne for 2.5 s. Its horizontal displacement is:

A. 4.8 m B. 12 m C. 30 m D. 49 m

10. A boat moves 4.0 m/s north relative to the water while the current moves 3.0 m/s east. Its velocity relative to shore is:

A. 1.0 m/s [N] B. 5.0 m/s [37° E of N] C. 5.0 m/s [53° E of N] D. 7.0 m/s [NE]

11. A ball is launched at 10.0 m/s, 30° above the horizontal, from a roof 12.0 m high. Its flight time to the ground is closest to:

A. 1.06 s B. 1.56 s C. 2.16 s D. 3.06 s

12. A velocity-time graph has straight segments from (0 s, 0 m/s) to (2 s, +6 m/s), then to (4 s, -2 m/s). The total distance travelled is:

A. 10 m B. 11 m C. 12 m D. 14 m

13. A pilot needs a ground velocity of 60 km/h north while the wind is 20 km/h east. The required air velocity is approximately:

A. 63 km/h [18° W of N] B. 63 km/h [18° E of N] C. 80 km/h [N] D. 40 km/h [N]

14. A vehicle's velocity changes from 10 m/s north to 10 m/s east in 2.0 s. Its average acceleration is:

A. 0 m/s² B. 5.0 m/s² [E] C. 7.1 m/s² [45° S of E] D. 10 m/s² [45° N of E]

15. A 4.0 m long car at 90 km/h overtakes a 200 m long train moving at 54 km/h in the same direction. From the instant the car's front reaches the train's rear until the car's rear clears the train's front, the time is:

A. 7.3 s B. 20.4 s C. 22.7 s D. 73.4 s

B. True or False (5 marks)

Write **T** or **F** for each statement. Each is worth 1 mark.

16. At the top of a projectile's flight, its vertical velocity is momentarily zero. _____

17. Average speed is total distance travelled divided by total elapsed time. _____

18. A negative acceleration always means that an object is slowing down. _____

19. Two projectiles launched from the same height with the same initial vertical velocity but different horizontal velocities land at the same time, if air resistance is negligible. _____

20. A boat aimed due north in an eastward current can land directly north of its starting point simply by increasing its northward speed. _____

C. Lab Analysis (10 marks)

21. A motion sensor records the eastward position of a cart released from rest. The readings are:

Time, t (s)	0	1.0	2.0	3.0	4.0
Position, x (m [E])	0.0	1.2	4.8	10.8	19.2

a. Determine the cart's displacement from 0 to 4.0 s. **(2)**

b. Determine its average velocity from 0 to 4.0 s. **(2)**

c. Sketch a position-time graph using all five points, with labelled axes and a smooth trend. **(2)**

d. Use the positions at 1.0 s and 3.0 s to estimate the instantaneous velocity at 2.0 s. Explain the method. **(2)**

e. The data follow approximately $x = \frac{1}{2}at^2$. Determine a and state what feature of your graph indicates non-uniform motion. **(2)**

D. Applications (30 marks)

22. A search robot travels 120 m east, then 50 m north, in a total time of 40 s. Use a labelled vector sketch or components.

a. Find its total distance travelled. **(2)**

b. Find its resultant displacement, including direction. **(3)**

c. Find its average speed. **(2)**

d. Find its average velocity, including direction. Explain briefly why the speed and velocity magnitudes differ. **(3)**

23. A rescue flare leaves a 20.0 m high platform at 18.0 m/s, 30.0° above the horizontal. Take east as +x and up as +y.

a. Determine its initial horizontal and vertical velocity components. **(2)**

b. Determine its flight time before it reaches level ground. **(2)**

c. Determine its horizontal displacement. **(2)**

d. Determine its velocity just before impact, as a magnitude and direction. **(4)**

24. A drone flies horizontally at 20.0 m/s east and 12.0 m/s north, 45.0 m above a level road. At the instant it releases a small package, it is directly above point P. A cart on the road moves at 5.0 m/s east. Find where the cart must be at the release instant, relative to P, for the package to land in it. Take east as +x and north as +y.

a. Determine the package's flight time. **(2)**

b. Determine the package's horizontal velocity relative to the cart as components. **(2)**

c. Determine the required initial east and north coordinates of the cart relative to P. **(3)**

d. Express that initial position as a distance and bearing from P. **(2)**

e. State the package's horizontal velocity relative to the ground immediately after release and explain why. **(1)**